EXPERIMENT - 6

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6.1.1] Incremented Date

A) ALGORITHM

1. Start.

2. Read day.

3. Read month.

4. Read year.

5. If year ≤ 0, print "Invalid Date" and Stop.

6. If month < 1 or month > 12, print "Invalid Date" and Stop.

7. If month is 1, 3, 5, 7, 8, 10, 12 then max_days = 31.

8. Else if month is 4, 6, 9, 11 then max_days = 30.

9. Else if month is 2:

    If (year % 400 == 0) OR (year % 4 == 0 AND year % 100 ≠ 0)

        max_days = 29

    Else

        max_days = 28

10. If day < 1 or day > max_days, print "Invalid Date" and Stop.

11. If day < max_days then day = day + 1.

12. Else

    day = 1

    If month == 12

        month = 1

        year = year + 1

    Else

        month = month + 1

13. Print day-month-year in format DD-MM-YYYY.

14. Stop.

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B) PYTHON CODE

```
day = int(input())

month = int(input())

year = int(input())


def is_leap(year):

    if (year % 400 == 0) or (year % 4 == 0 and year % 100 != 0):

        return True

    return False


def days_in_month(month, year):

    if month in [1, 3, 5, 7, 8, 10, 12]:

        return 31

    elif month in [4, 6, 9, 11]:

        return 30

    elif month == 2:

        return 29 if is_leap(year) else 28

    else:

        return 0


if year <= 0 or month < 1 or month > 12:

    print("Invalid Date")

elif day < 1 or day > days_in_month(month, year):

    print("Invalid Date")

else:

    day += 1

    if day > days_in_month(month, year):

        day = 1

        month += 1

        if month > 12:

            month = 1

            year += 1

    print(f"{day:02d}-{month:02d}-{year}")
```

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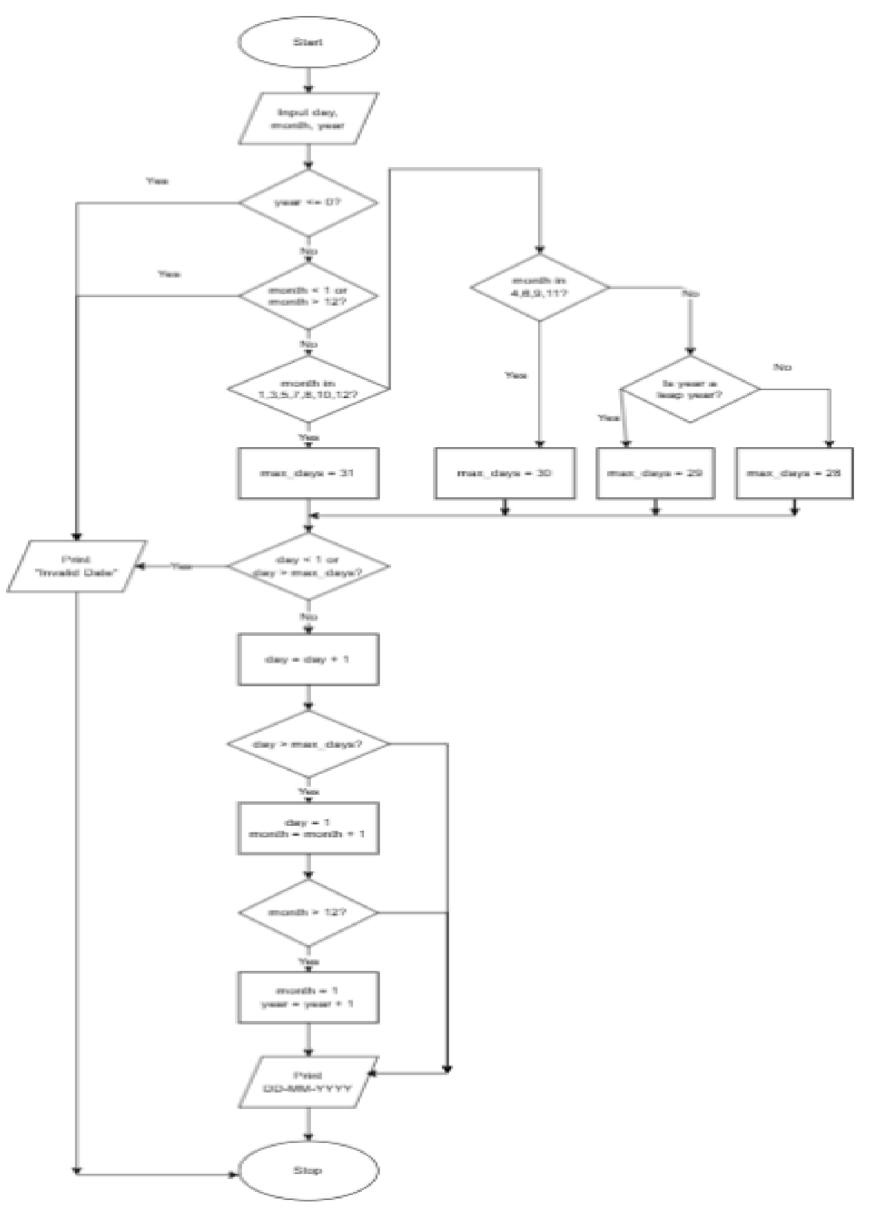
C) FLOWCHART



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D) EXCECUTION